Statistical Methods for Computer Science

Exercises: Probability

1) Which of the following is an example of a random experiment?
A) A customer inspects a delivery and finds an error.
B) Marketing budgets a 10% increase in advertising.
C) The post office misplaces a letter.
D) A student increases her average study time/week and improves her grades.

2) The set of all possible outcomes from a random experiment is called the sample:
A) population.
B) space.
C) probability.
D) event.

3) Which of the following is not an example of a random experiment?
A) daily change of stock market index prices
B) customer makes a purchase or not
C) a survey to rate quality of service
D) coin-toss, heads or tails

4) The possible outcomes from a random experiment are called:
A) sample space.
B) parameters.
C) basic groups.
D) basic outcomes.

5) A subset of outcomes is defined as a(n):
A) sample.
B) event.
C) outcome.
D) probability.

6) When events have no common basic outcomes, they are said to be:
A) mutually exclusive.
B) mutually related.
C) mutually apart.
D) collectively exhaustive.

7) If the union of several events covers the entire sample space, it is said these events are:
A) mutually exclusive.
B) mutually related.
C) mutually apart.
D) collectively exhaustive.

8) The proportion of times that an event will occur, assuming that all outcomes in a sample space are equally likely to occur, is called:
A) objective probability.
B) classical probability.
C) relative frequency probability.
D) subjective probability.

9) ________ probability is the number of events in the population that meet the condition divided by the total number in the population.
A) Objective
B) Classical
C) Relative frequency
D) Subjective

10) The expression of an individual's degree of belief about the chance that an event will occur is called ________ probability.
A) objective
B) classical
C) relative frequency
D) subjective

<u>THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:</u>
Suppose you roll a pair of dice. Let *A* be the event that you observe an even number. Let *B* be the event that you observe a number greater than seven.

11) What is the intersection of events *A* and *B*?
A) [8, 10, 12]
B) [7, 8, 9, 10, 11, 12]
C) [2, 4, 6, 8, 10, 12]
D) [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]

12) What is the union of events *A* and *B*?
A) [8, 9, 10, 11, 12]
B) [8, 10, 12]
C) [2, 4, 6, 8, 9, 10, 11, 12]
D) [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]

13) What is the complement of event *A*?
A) [8, 9, 10, 11, 12]
B) [3, 5, 7, 9, 11]
C) [1, 5, 10, 11, 12]
D) [8, 10, 12]

14) What is the complement of event *B*?
A) [3, 5, 7, 9, 11]
B) [2, 4, 6, 8, 10, 12]
C) [2, 3, 4, 5, 6, 7]
D) [7, 8, 9, 10, 11, 12]

15) What is $\overline{A} \cap B$?
A) [9, 11]
B) [2, 3, 4, 6]
C) [5, 7, 8, 10, 12]
D) [3, 5]

16) What is $A \cap \overline{B}$?
A) [9, 11]
B) [8, 10, 12]
C) [3, 5, 7]
D) [2, 4, 6]

17) What is $\overline{A} \cap \overline{B}$?
A) [2, 4, 6]
B) [3, 5, 7]
C) [8, 9, 10, 11, 12]
D) [$\varnothing$]

18) What is $\overline{A} \cup \overline{B}$?
A) [2, 3, 4, 5, 6, 7]
B) [3, 5, 7]
C) [2, 3, 4, 5, 6, 7, 9, 11]
D) [2, 4, 6]

19) What is $\overline{A} \cup B$?
A) [4, 5, 7, 8, 11, 12]
B) [8, 9, 10, 11, 12]

C) [2, 3, 4, 5, 6, 7]
D) [3, 5, 7, 8, 9, 10, 11, 12]

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:
The production manager at a local manufacturing plant is concerned about work stoppages in the four production lines. In particular, the manager is evaluating the likelihood of the next stoppage occurring on production line 1.

30) If each of the lines is equally likely to have the next stoppage, the probability that production line 1 is stopped next is 25%. This is an example of:
A) subjective probability.
B) classical probability.
C) relative frequency probability.

31) Based on previous stoppage records, the manager figures that the probability that production line 1 is stopped next is 25%. This is an example of:
A) subjective probability.
B) classical probability.
C) relative frequency probability.

32) Based on his knowledge of the causes of stoppages, the manager states that the probability that production line 1 is stopped next is 25%. This is an example of:
A) subjective probability.
B) classical probability.
C) relative frequency probability.

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:
In a furniture manufacturing plant, a customer survey indicates that blemishes in the finish are a major concern. The table shown below displays a quality manager's probability assessment of the number of defects in the finish of new furniture.

Number of Defects	0	1	2	3	4	5
Probability	0.34	0.25	0.19	0.11	0.07	0.04

33) Let event *A* be that there are more than three defects and let event *B* be that there are four or fewer defects. Which of the following statements is true?
A) $P(A \cap B) = 0.18$
B) $P(A \cup B) = 0.07$
C) Events *A* and *B* are collectively exhaustive.
D) Events *A* and *B* are mutually exclusive.

34) Let event *A* be that there are more than two defects and let event *B* be that there are four or fewer defects. Which of the following statements is true?
A) $P(A \cap B) = 0.18$
B) $P(A \cup B) = 0.07$
C) $P(\overline{A}) = 0.58$
D) $P(\overline{B}) = 0.89$

35) Let *A* be the event that there is at least one defect and let event *B* be that there is at most three defects. Which of the following statements is true?
A) $P(A \cap B) = 0.55$
B) $P(A \cup B) = 0.96$
C) $P(A) = 0.34$
D) $P(\overline{B}) = 0.89$

36) A gumball machine has five different colored gumballs: red, blue, white, green, and yellow. If you buy three gumballs, how many different combinations of colors could you buy?

A) 60
B) 20
C) 10
D) 6

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:
Ted's Surfboard Shop makes surfboards by hand. The number of surfboards that Ted makes in a week depends on the wave conditions. Ted has estimated the following probabilities for surfboard production for the next week.

Number of Surfboards	5	6	7	8	9	10
Probability	0.13	0.22	0.31	0.17	0.13	0.04

37) Let event *A* be that Ted produces more than seven surfboards and let event *B* be that he produces exactly six surfboards. Which of the following statements is true?
A) $P(A \cap B)$=0.31
B) events *A* and *B* are collectively exhaustive
C) $P(\overline{A} \cup \overline{B})$=0.44
D) events *A* and *B* are mutually exclusive

38) Let event *A* be that Ted produces more than six surfboards, and let event *B* be that he produces less than eight surfboards. Which of the following statements is true?
A) $P(A \cap B)$ = 0.75
B) Events *A* and *B* are collectively exhaustive.
C) $P(\overline{A})$ = 0.53
D) Events *A* and *B* are mutually exclusive.

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:
The probability that interest rates on housing loans will go up in the next 6 months is estimated to be 0.20. The probability that house sales will decrease is estimated to be 0.6. The probability that interest rates will go up and house sales will decrease is estimated to be 0.15.

39) The probability that interest rates increase and house sales decrease is:
A) 0.75
B) 0.85
C) 0.71
D) 0.15

40) The probability of an increase in interest rates and not a decrease in house sales is:
A) 0.20
B) 0.05
C) 0.25
D) 0.50

41) The probability of a decrease in house sales and not an increase in interest rates is:

A) 0.45
B) 0.25
C) 0.20
D) 0.05

42) The probability of not an increase in interest rates and not a decrease in house sales is:
A) 0.50
B) 0.35
C) 0.20
D) 0.05

43) The probability that house sales will go down given that interest rates will go up is:
A) 0.95
B) 0.90
C) 0.75
D) 0.80

<u>THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:</u>
A multiple choice quiz has five questions, each with five answers, *A* through *E*. Assume you just guess on all of the questions.

44) What is the probability that you guessed on all five questions right?
A) 0.00032
B) 0.03125
C) 0.20
D) 0.50

45) What is the probability that you get exactly three questions right?
A) 0.00032
B) 0.0016
C) 0.008
D) 0.04

46) A research project on retention at a large university indicated that 20% of the students had a problem with stress while 25% reported problems with financial resources and 15% had a problem with both. What is the probability that a particular student has a problem with either stress or financial resources?
A) 0.15
B) 0.30
C) 0.45
D) 0.60

47) The probability that an employee at a company uses illegal drugs is 0.08. The probability than an employee is male is 0.55. Assuming that these events are independent, what is the probability that a randomly chosen employee is a male drug user?
A) 0.742
B) 0.145
C) 0.044
D) 0.006

48) A survey of executives revealed that 35% of them regularly read The Wall Street Journal, 20% read Forbes, and 10% read both The Wall Street Journal and Forbes. What is the probability that a particular

executive reads either The Wall Street Journal or Forbes?
A) 0.45
B) 0.35
C) 0.55
D) 0.65

49) In a longitudinal economic development study, market research indicated that the odds of a new business succeeding after five years are 1 to 9. That means that the probability of a business actually succeeding is:
A) 0.11
B) 0.10
C) 0.09
D) 0.08

50) A junior executive looking at his business attire in his closet notes that he has five suits, six shirts, and three pairs of shoes. He is going on a business trip and needs to take two of each. How many different combinations of outfits could he take?
A) 680
B) 320
C) 450
D) 224

<u>THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:</u>
A manufacturer of automobiles conducted a market survey. Eighty percent of the customers want better fuel efficiency, while 55% want a vehicle navigation system and 45% percent want both features.

54) The probability that a person wants better fuel efficiency and also a vehicle navigation system is:
A) 0.90
B) 0.10
C) 0.45
D) 0.35

55) The probability that a person wants better fuel efficiency but not a vehicle navigation system is:
A) 0.90
B) 0.10
C) 0.45
D) 0.35

56) The probability that a person wants a vehicle navigation system but not better fuel efficiency is:
A) 0.90
B) 0.10
C) 0.45
D) 0.35

57) The probability that a person wants neither better fuel efficiency nor a vehicle navigation system is:
A) 0.90
B) 0.10
C) 0.45
D) 0.35

58) The probability that a person wants either better fuel efficiency or a vehicle navigation system is:
A) 0.90

B) 0.10
C) 0.45
D) 0.35

59) The probability that a person does not want a vehicle navigation system is:
A) 0.45
B) 0.10
C) 0.90
D) 0.35

60) The probability that a person wants a better fuel efficiency given that he wants a vehicle navigation system is approximately:
A) 0.90
B) 0.45
C) 0.82
D) 0.41

61) The probability that a person wants a vehicle navigation system given that he wants a better fuel efficiency is approximately:
A) 0.80
B) 0.40
C) 0.45
D) 0.56

62) A junior executive looking at his business attire in his closet notes that he has eight suits, six shirts, and four pairs of shoes. He is going on a business trip and needs to take two of each. How many different combinations of outfits could he take?
A) 2240
B) 3320
C) 1680
D) 2520

63) The purchasing agent for a municipality has contracted with a local car dealer to purchase four cars. The dealer has 25 cars on his lot; 10 red, 7 blue, 6 white, and 2 purple. If the purchasing agent has no control over the colors he receives, what is the probability that he receives at least one of the purple cars?
A) 0.33
B) 0.30
C) 0.36
D) 0.39

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:
A recent survey showed that 15% of computer programmers have experienced some form of wrist pain from typing, and that 25% are taking aspirin daily. Six percent of all programmers have both experienced some form of wrist pain from typing and taken aspirin on a daily basis.

64) What is the probability that a programmer who has wrist pain takes aspirin on a daily basis?
A) 0.40
B) 0.24
C) 0.57
D) 0.66

65) What is the probability that a programmer who takes aspirin on a daily basis has wrist pain?
A) 0.09

B) 0.19
C) 0.24
D) 0.40

66) What is the probability that a programmer has wrist pain and does not take aspirin on a daily basis?
A) 0.25
B) 0.15
C) 0.19
D) 0.09

67) What is the probability that a programmer does not have wrist pain and does not take aspirin on a daily basis?
A) 0.75
B) 0.66
C) 0.85
D) 0.25

68) What is the probability that a programmer has wrist pain or takes aspirin on a daily basis?
A) 0.66
B) 0.25
C) 0.34
D) 0.15

69) As the office manager for a medical office with ten doctors, you are responsible for developing the roster for the on-call shift for the next three nights. How many different ways could you assign a different doctor to each of the next three nights?
A) 1020
B) 840
C) 720
D) 640

70) Which of the following statements is true?
A) If events *A* and *B* are complements, then the intersection of *A* and the complement of *B* is the sample space.
B) If events *A* and *B* are mutually exclusive, then the intersection of *A* and *B* is the empty set.
C) If events *A* and *B* are mutually exclusive, then the union of *A* and *B* is the sample space.
D) If events *A* and *B* are mutually exclusive, then the union of *A* and *B* is the empty set.

71) In a recent article it was reported that 35.4% of all high school students smoke cigarettes. 65% of these students plan on going to college. What is the probability that a randomly selected student smokes cigarettes and plans on going to college?
A) 0.354
B) 0.230
C) 0.124
D) 0.412

72) An office of six people is plagued by high absenteeism. It is thought that the probability that an employee is absent on a particular day is 0.03. Assuming that the event that one person is absent on a particular day is independent of the absence of any other employee, what is the probability that at least one employee is absent tomorrow?
A) 0.121
B) 0.180
C) 0.150
D) 0.167

73) A gumball machine has six red gumballs, four blue gumballs, and three yellow gumballs. If you buy three gumballs, what is the probability that you get three different colors?
A) 0.252
B) 0.167
C) 0.125
D) 0.244

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:
Given A= {E_3, E_4, E_8, E_{10}} and B = {E_3, E_4, E_6, E_9}

79) What is the intersection of *A* and *B*?
A) E_3, E_4
B) E_3, E_4, E_6, E_8, E_9, E_{10}
C) E_1, E_2, E_5, E_7
D) E_6, E_8, E_9, E_{10}

80) What is the union of *A* and *B*?
A) E_3, E_4
B) E_3, E_4, E_6, E_8, E_9, E_{10}
C) E_1, E_2, E_5, E_7
D) E_6, E_8, E_9, E_{10}

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:
A supplier is evaluating a firm to manufacture a subassembly. Quality data from past inspections reveal the following probabilities for number of defective parts in a shipment:

Number Defective	0	1	2	>2
Probability	0.80	0.10	0.06	0.04

81) What is the probability that there will be fewer than 2 defective parts in a shipment?
A) 0.20
B) 0.90
C) 0.86
D) 0.10

82) What is the probability that the shipment will have defective parts?
A) 0.20
B) 0.90
C) 0.86
D) 0.10

83) What is the probability that the shipment will have at least two defective parts?
A) 0.20
B) 0.90
C) 0.86
D) 0.10

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:

In a survey of top executives, it was found that 46% had either traveled internationally on business or could fluently speak a foreign language. The probability that an executive who had not traveled internationally could speak a foreign language was 10%. It was found that only 4% of the executives had traveled internationally and could speak a foreign language.

263) What is the probability that an executive does not speak a foreign language fluently?

264) What is the probability that an executive has traveled internationally and could not speak a foreign language fluently?

265) What is the probability that an executive has not traveled internationally and could speak a foreign language fluently?

266) Firms are increasingly asking applicants to submit to drug tests. Suppose that drug tests can identify a drug user 98% of the time. However, 1% of the time the test indicates a positive result although the applicant is not a drug user. If 10% of all applicants are drug users, what is the probability that a person who has tested positive for drug use is not really a drug user?

<u>THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:</u>
In a recent survey about capital punishment, 64% of the respondents said that they support capital punishment. Females comprised 48% of the sample, and of the females, 46% supported capital punishment.

267) What is the probability that a randomly selected person is a female and capital punishment supporter?

268) Suppose we select a capital punishment supporter so that we can ask some follow up questions. What is the probability that the person we select is female?

269) What is the probability that a randomly selected person is a male and does support capital punishment?

270) What is the probability that a randomly selected person is a male and does not support capital punishment?

271) What is the probability that a randomly selected person is a female and does not support capital punishment?

272) Suppose we select a person who does not support capital punishment, what is the probability that the person is male?

273) Suppose we select a female respondent, what is the probability that she is not a capital punishment supporter?

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:
Consider a sample space defined by events A_1, A_2, B_1, B_2. Let $P(A_1) = 0.40$, $P(B_1 \mid A_1) = 0.60$ and $P(B_1 \mid A_2) = 0.70$

274) What is $P(A_2)$?

275) What is $P(A_1 \cap B_1)$?

276) What is $P(A_1 \cap B_2)$?

277) What is $P(A_2 \cap B_1)$?

278) What is $P(B_1)$?

279) What is $P(B_2)$?

280) What is $P(A_2 \cap B_2)$?

281) What is $P(A_1 \mid B_1)$?

282) What is $P(A_1 \mid B_2)$?

283) What is $P(A_2 \mid B_1)$?

284) What is $P(A_2 \mid B_2)$?

285) What is $P(B_2 \mid A_1)$?

286) What is $P(B_2 \mid A_2)$?